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**Department of Information Technology**

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Student Name: Vishal R Mahajan Student Roll No: 1

# Experiment – 5 : Storage as a Service

**Aim:** To study and Implement Storage as a Service using different cloud platform AWS / GC / AZURE (Free Tier Cloud Platforms).

**Objective:** After performing the experiment, the students will be able to –

- Perform Amazon S3 Operations (Create/delete a bucket, Write an object, Read an object, Delete an object)
- Bucket versioning
- Know how to enable static website hosting on Amazon S3
- Know how to use lifecycle configuration rules

**Lab objective mapped : ITDO802.3** : To learn, simulate and examine cloud service model and deploy IAAS, SAAS, PAAS services over different cloud platforms.

**Prerequisite:** Concept of Operating System, Infrastructure.

**Requirements:** Cloud Login**,** Desktop, Browser, Internet etc.

**Pre-Experiment Theory:**

**CLOUD Storage services (describe-soft copy form)**

**S3 (describe-soft copy form)**

**Azure Blob Storage (describe-soft copy form)**

**Procedure**

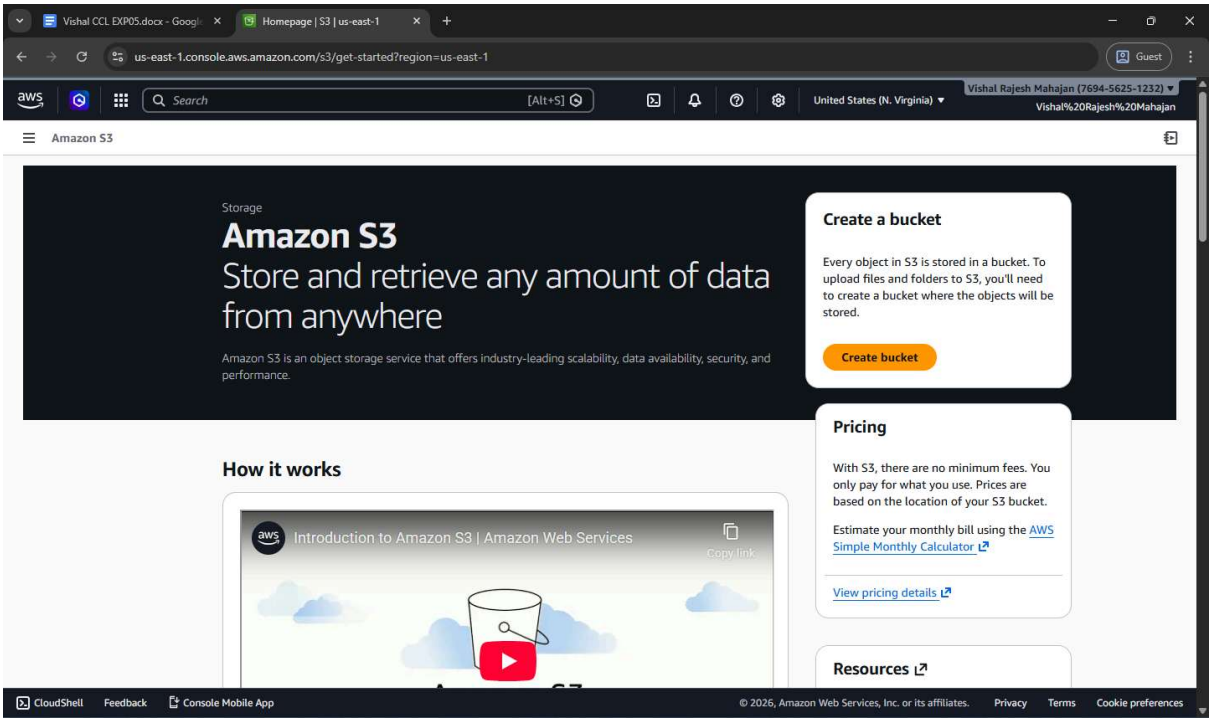
**Create an Amazon Simple Storage Service (Amazon S3) Bucket in your selected region.**

1. Log in to the AWS Management Console.

2. Choose an appropriate region.

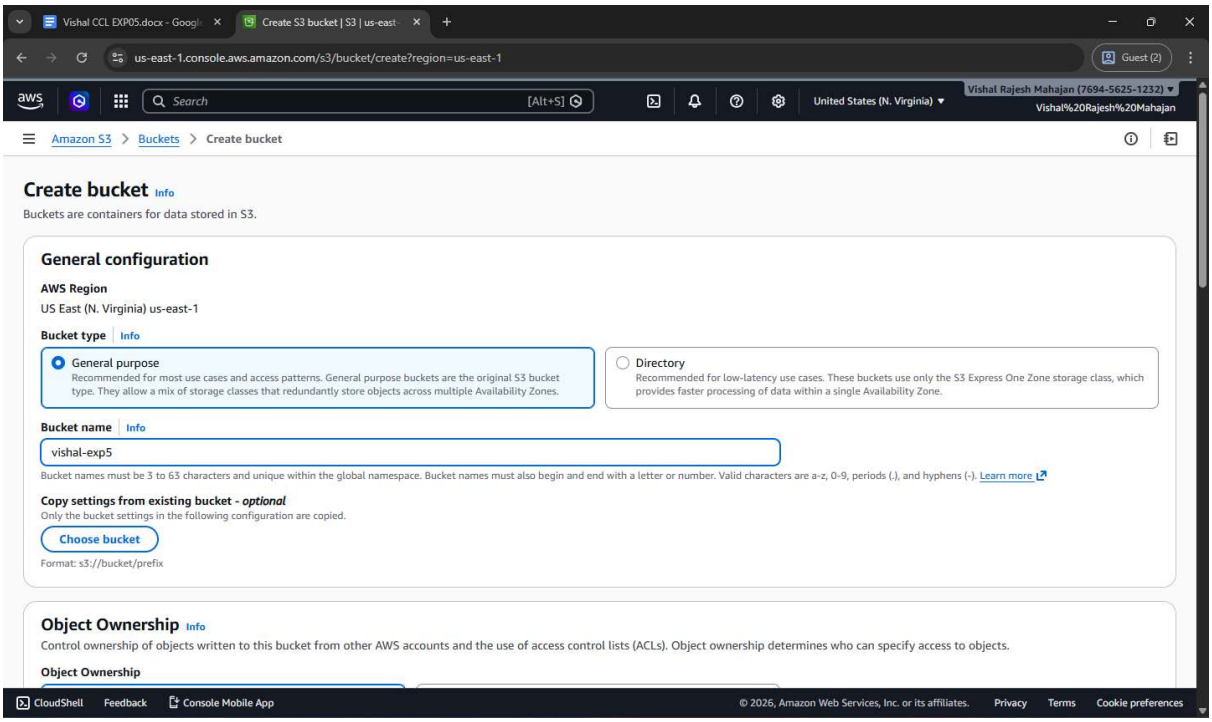
3. Navigate to the Amazon S3 console. Notice that the region indicator now says

Global.




4. Start the create bucket process.

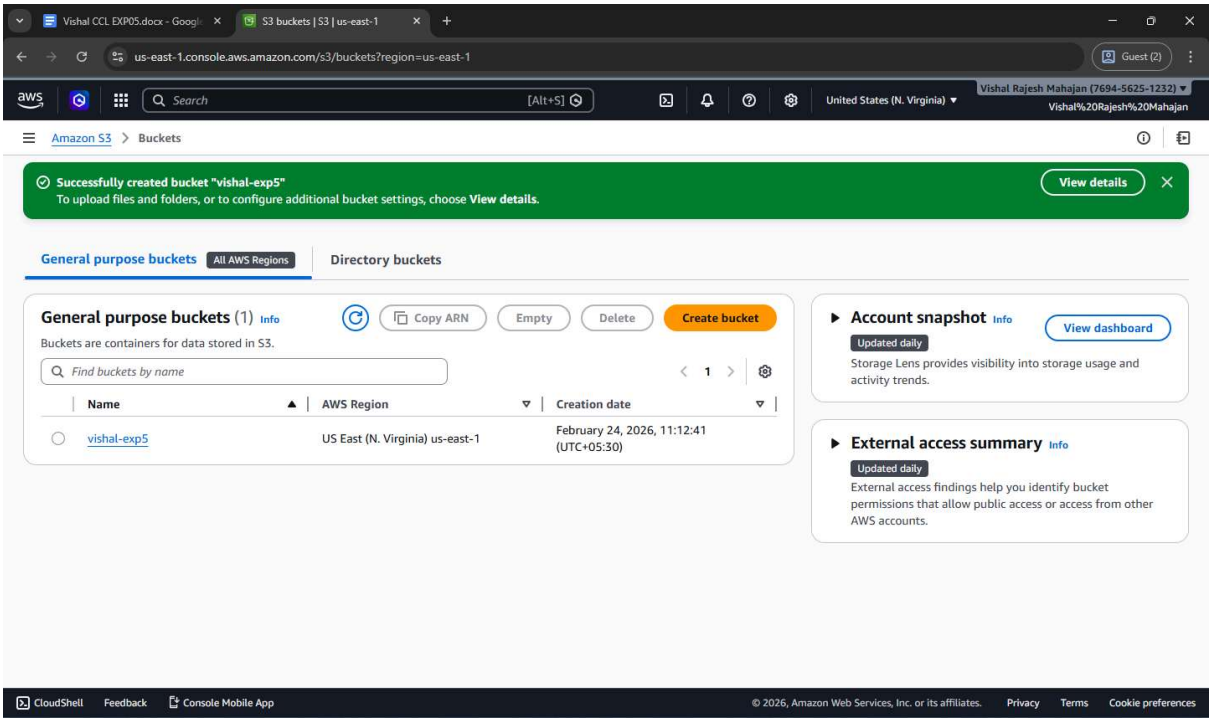
5. When prompted for Bucket Name, use unique name.




6. Choose a region.

7. Try to create the bucket.

8.You should now have a new Amazon S3 bucket.

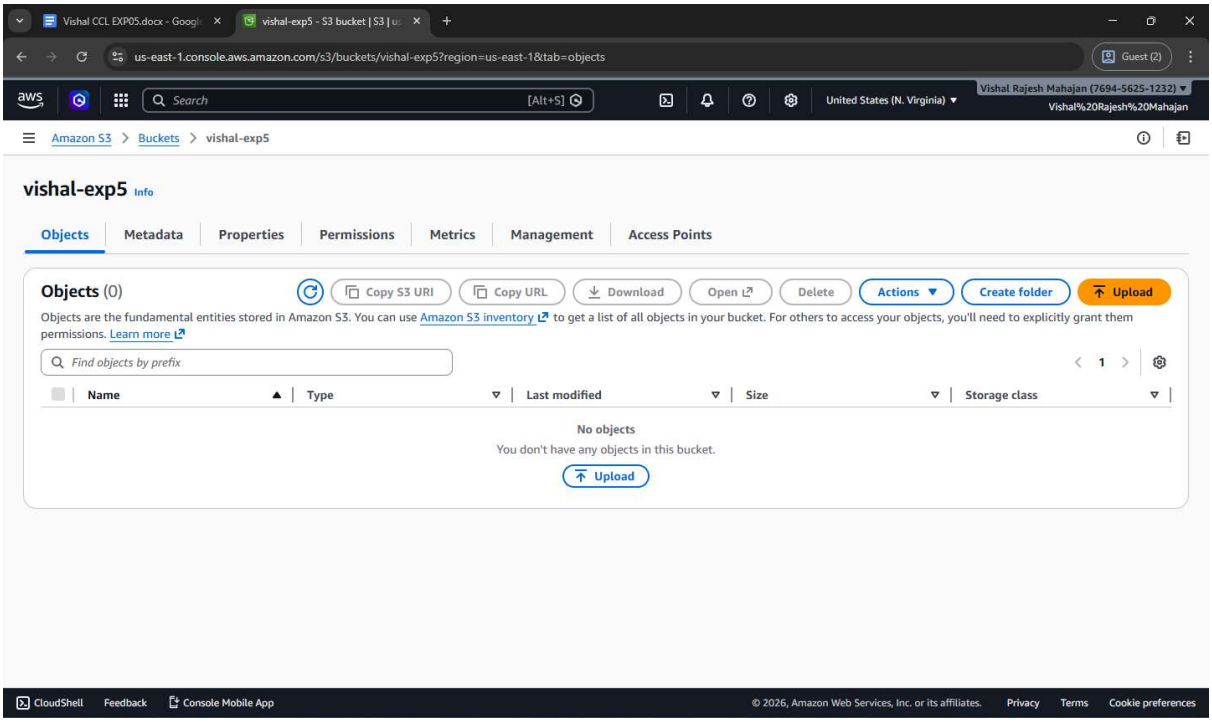


**Upload, Make Public, Rename, and Delete Objects in Your Bucket**

Upload an Object

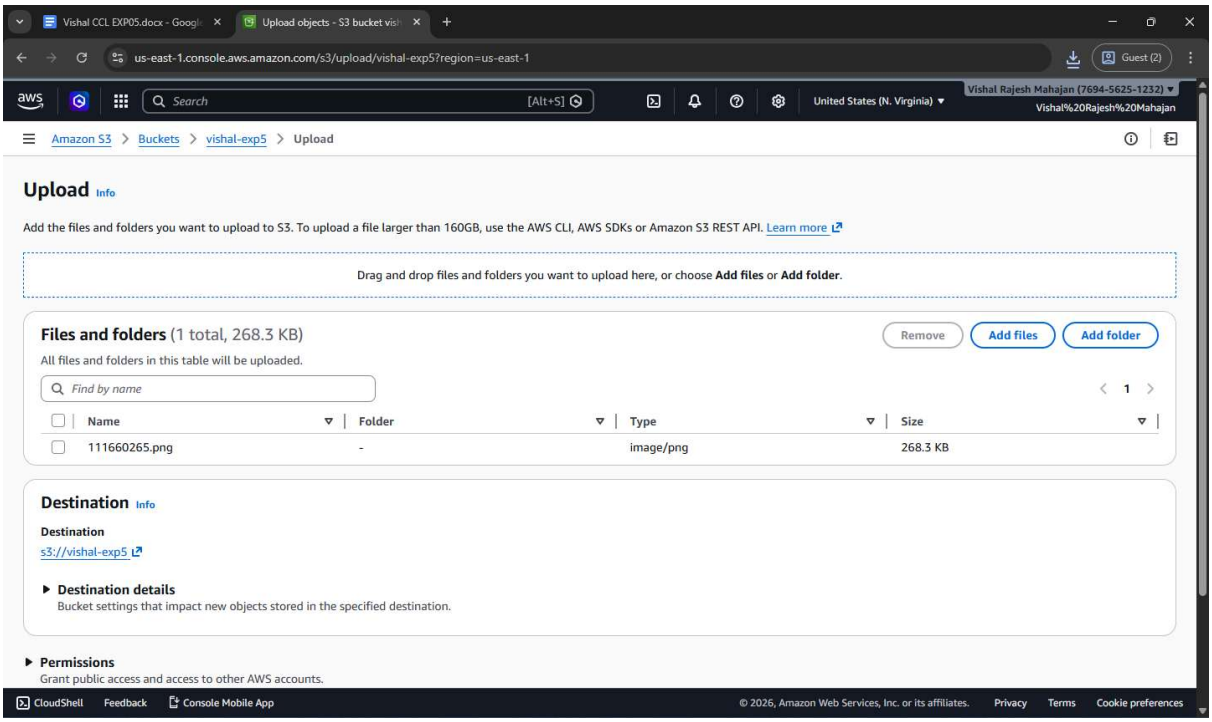
1. Load your new bucket in the Amazon S3 console.

2. Select Upload, then Add Files.

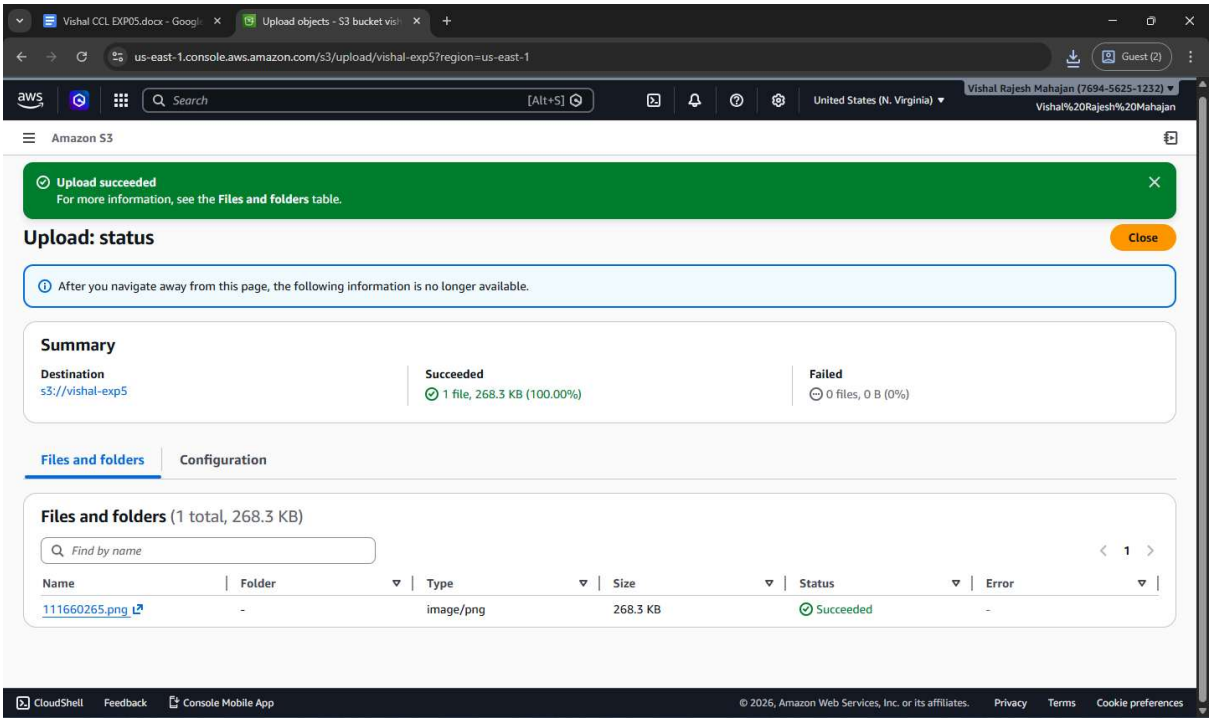



3. Locate a file on your PC.

4. Select a suitable file, then Start Upload. You will see the status of your file in the Transfers section.




5. After your file is uploaded, the status should change to Done.

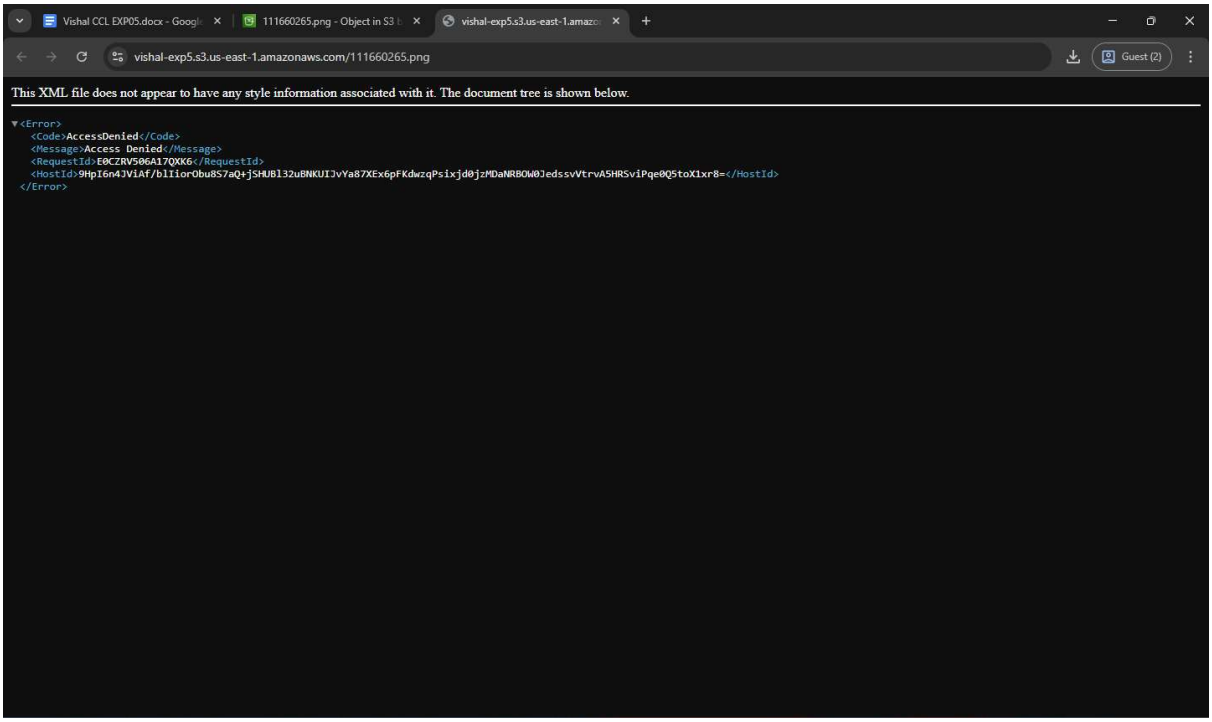


Open the Amazon S3 URL

6. Now open the properties for the object. The properties should include bucket, name, and link.

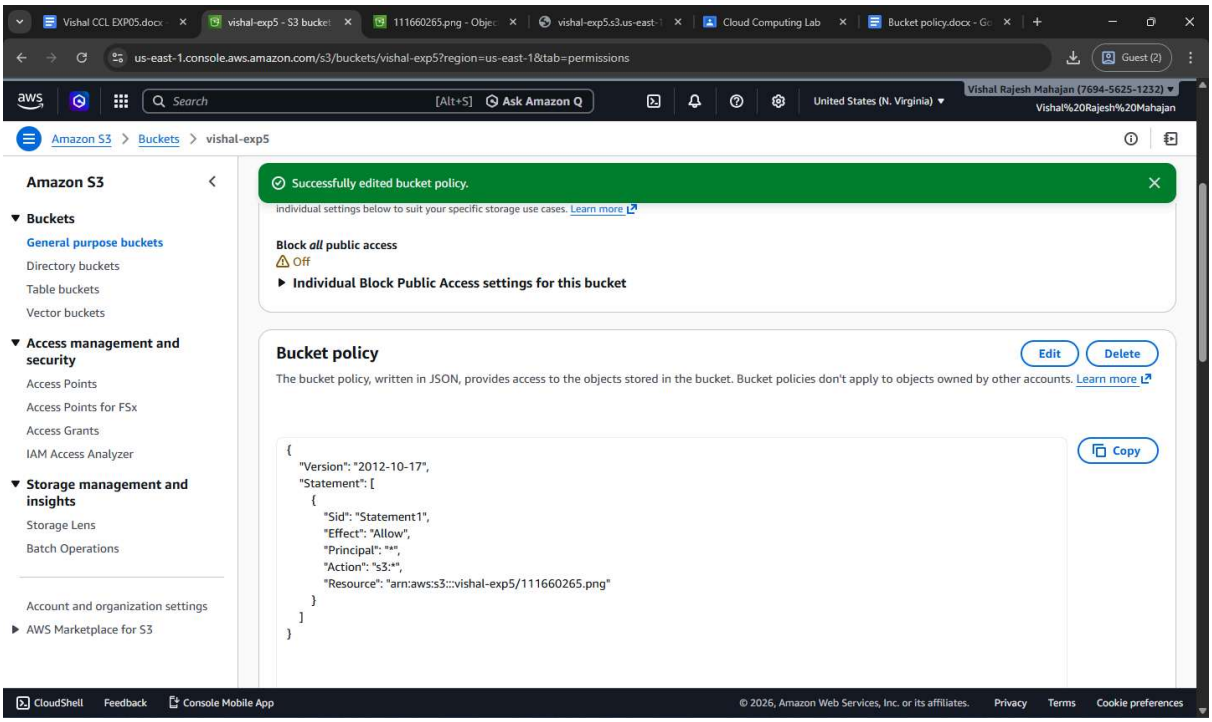
7. Copy the Amazon S3 URL for the object.

8. Paste the URL in the address bar of a new browser window or tab. You should get a message with an XML error code AccessDenied. Even though the object has a URL, it is private by default, so it cannot be accessed by a web browser.

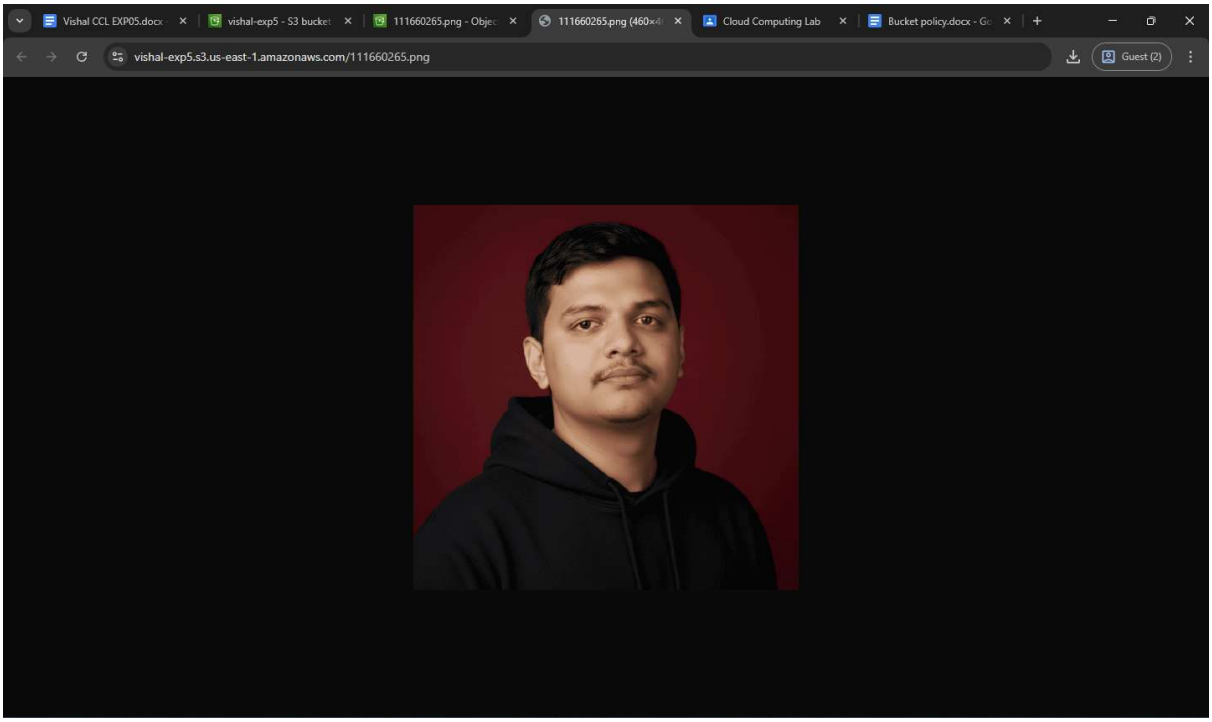



Make the Object Public

9. Go back to the Amazon S3 Console and select Make Public.



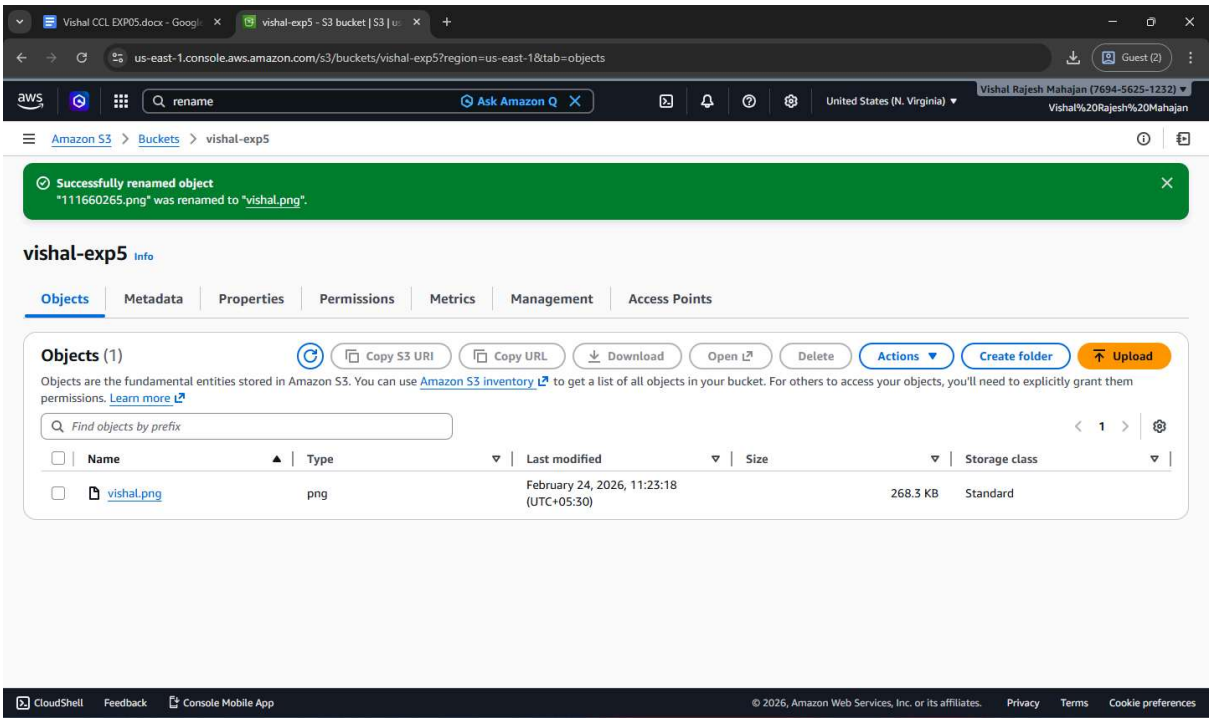

10. Copy the Amazon S3 URL again and try to open it in a browser or tab. Your public image file should now display in the browser or browser tab.



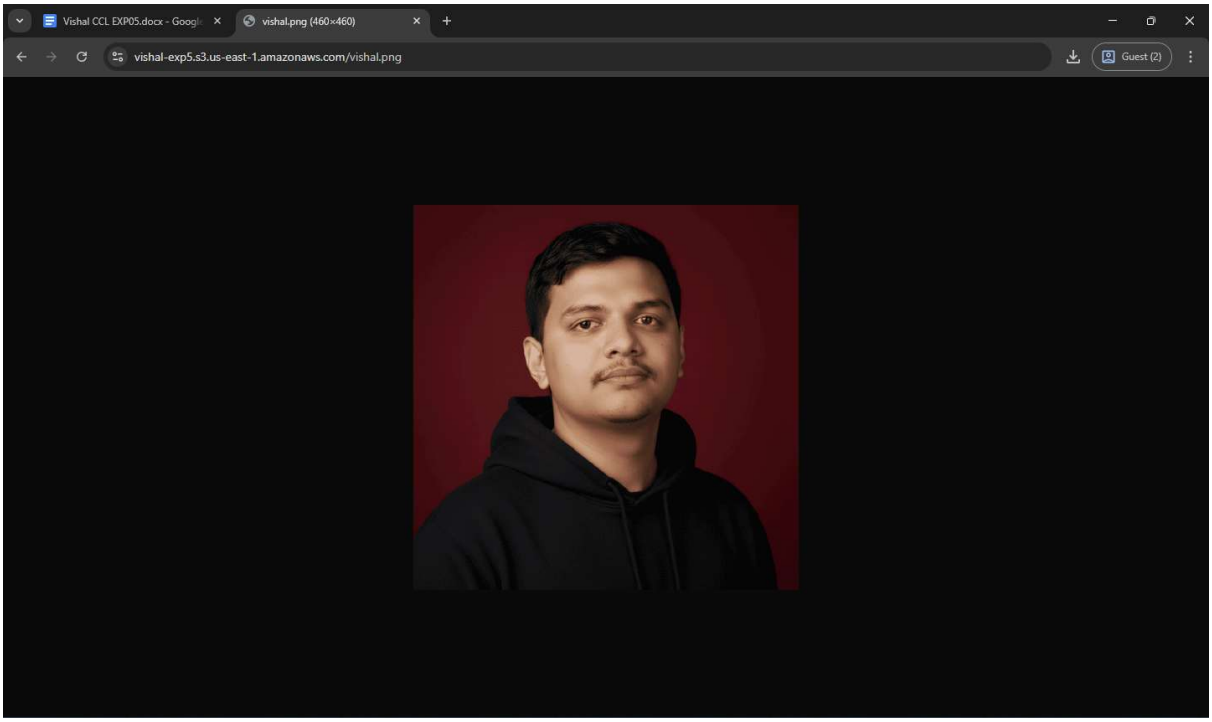
Rename Object

11. In the Amazon S3 console, select Rename.

12. Rename the object, but keep the same file extension.

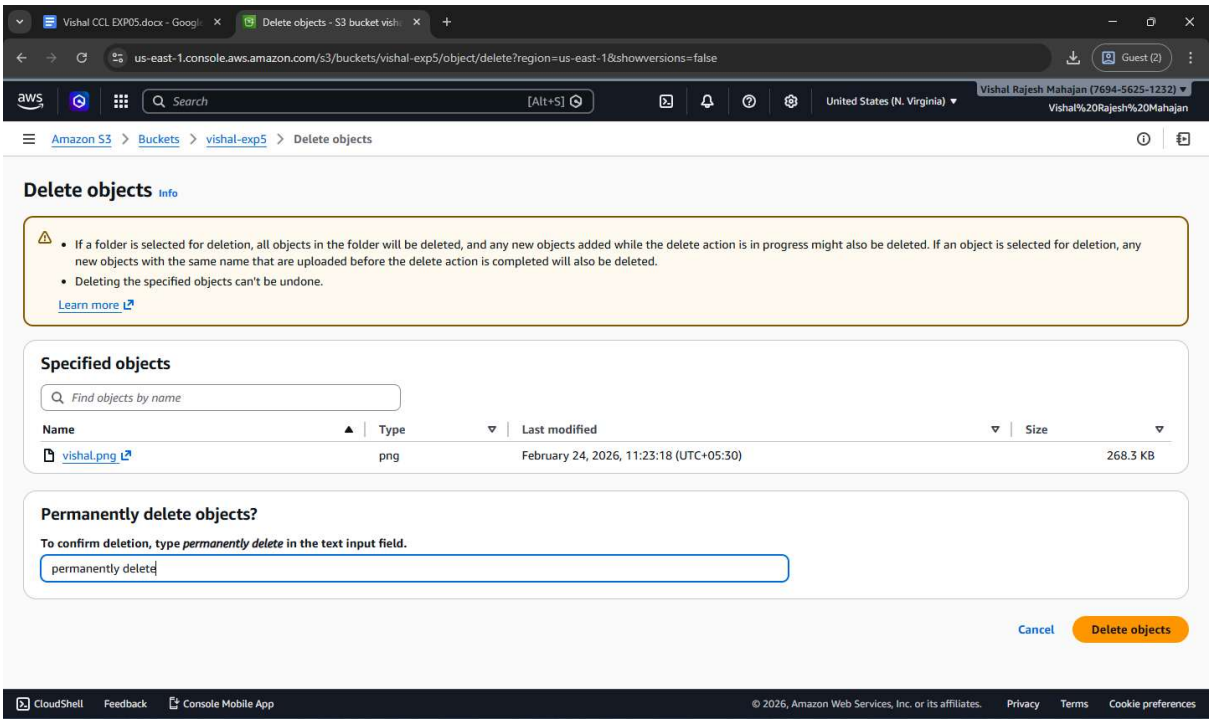



13. Copy the new Amazon S3 URL and try to open it in a browser or tab. You should see the same image file.



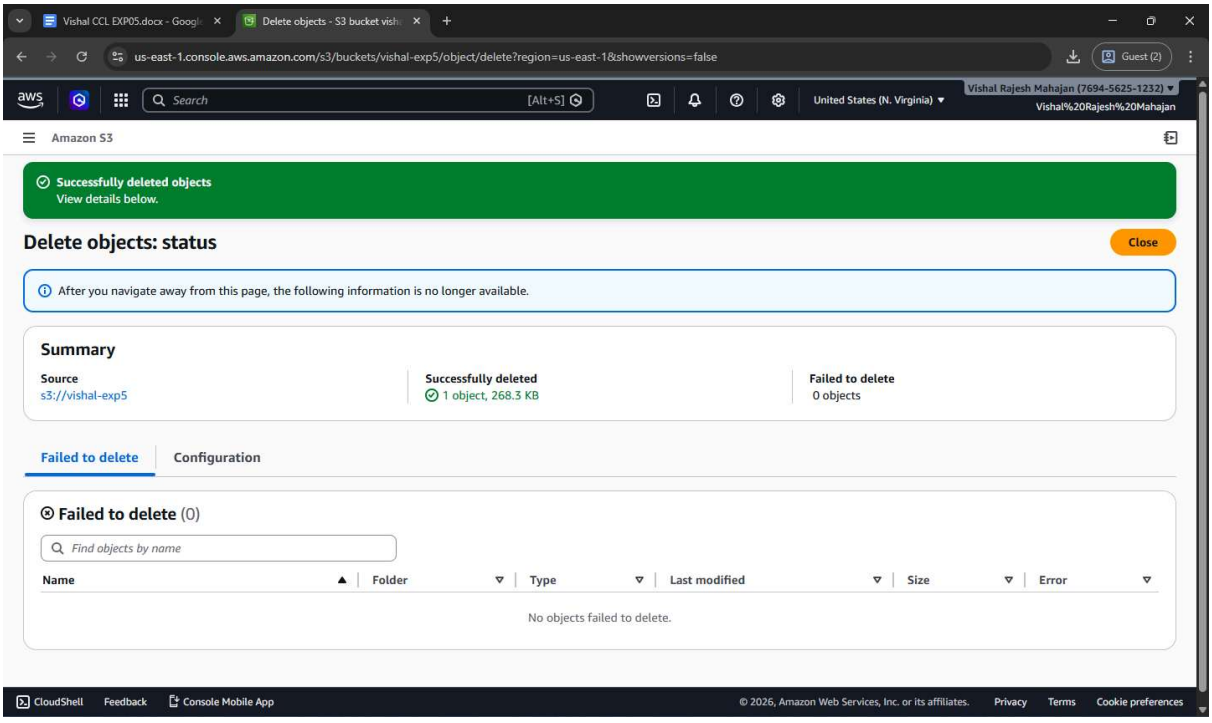
## Delete the Object

14. In the Amazon S3 console, select Delete. Select OK when prompted if you want to delete the object.




15. The object has now been deleted.

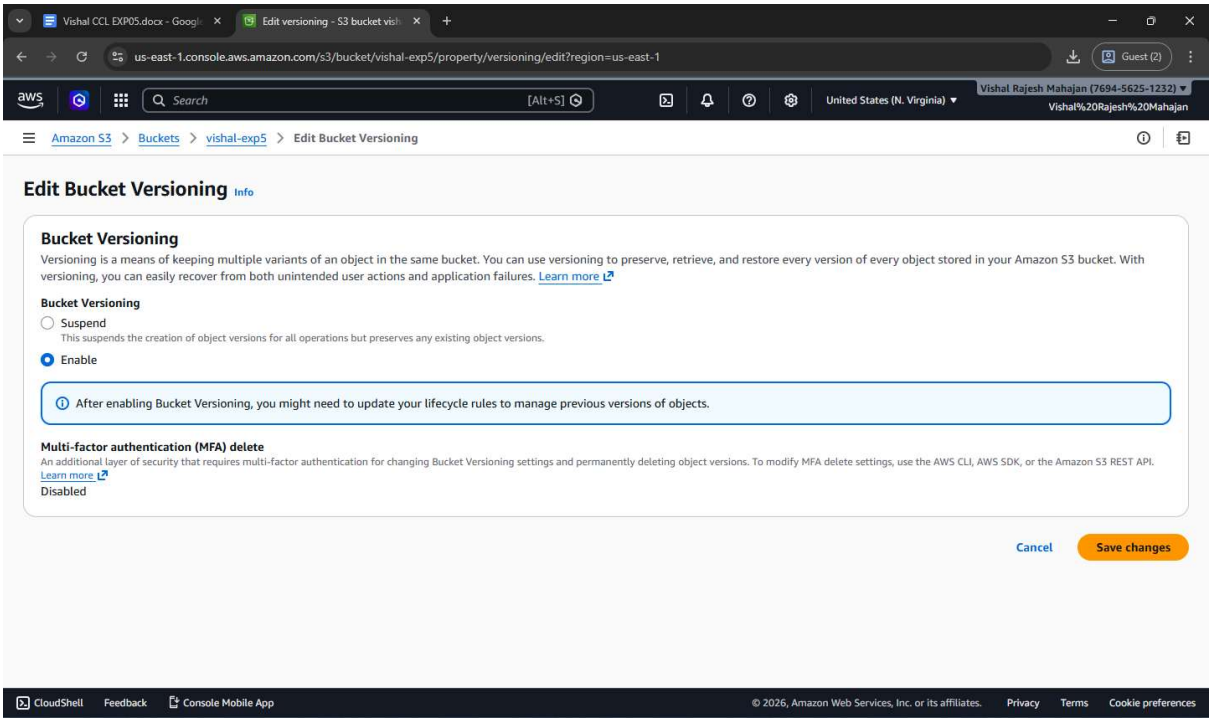
16. To verify, try to reload the deleted object's Amazon S3 URL. You should once again get the XML AccessDenied error message.

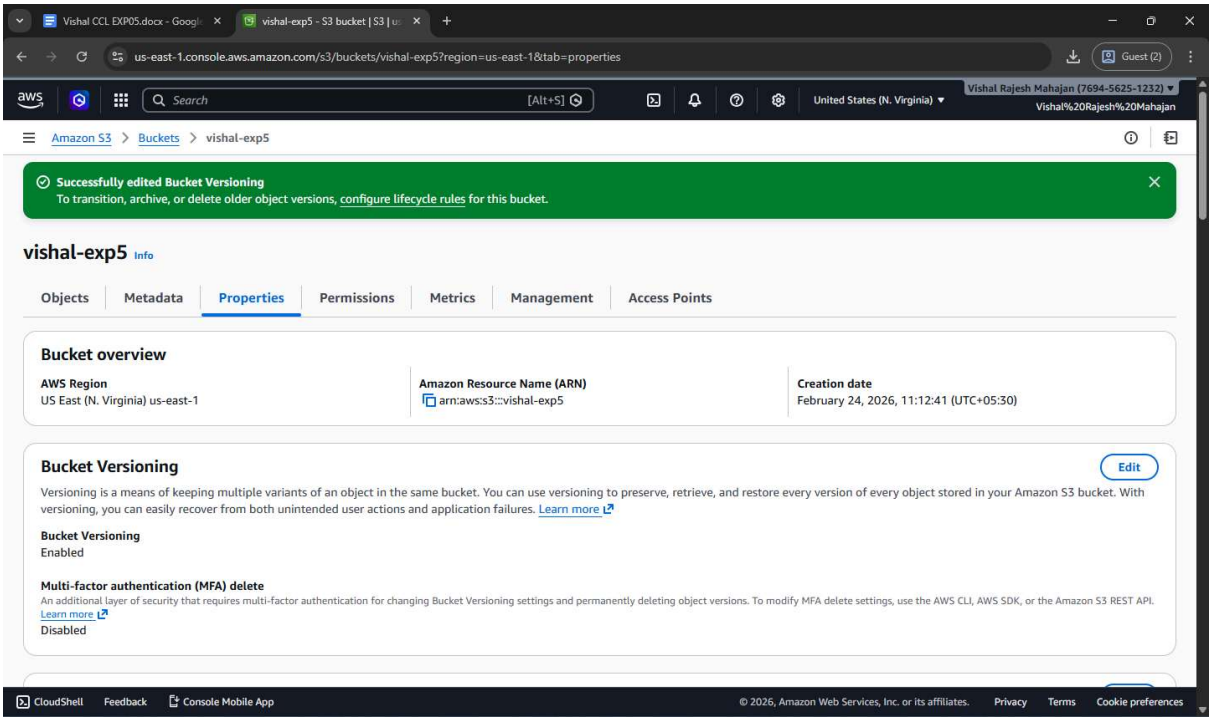


## Enable Version Control

1. In the Amazon S3 console, load the properties of your bucket. Don't open the bucket.

2. Enable versioning in the properties and select OK to verify. Your bucket now has versioning enabled.



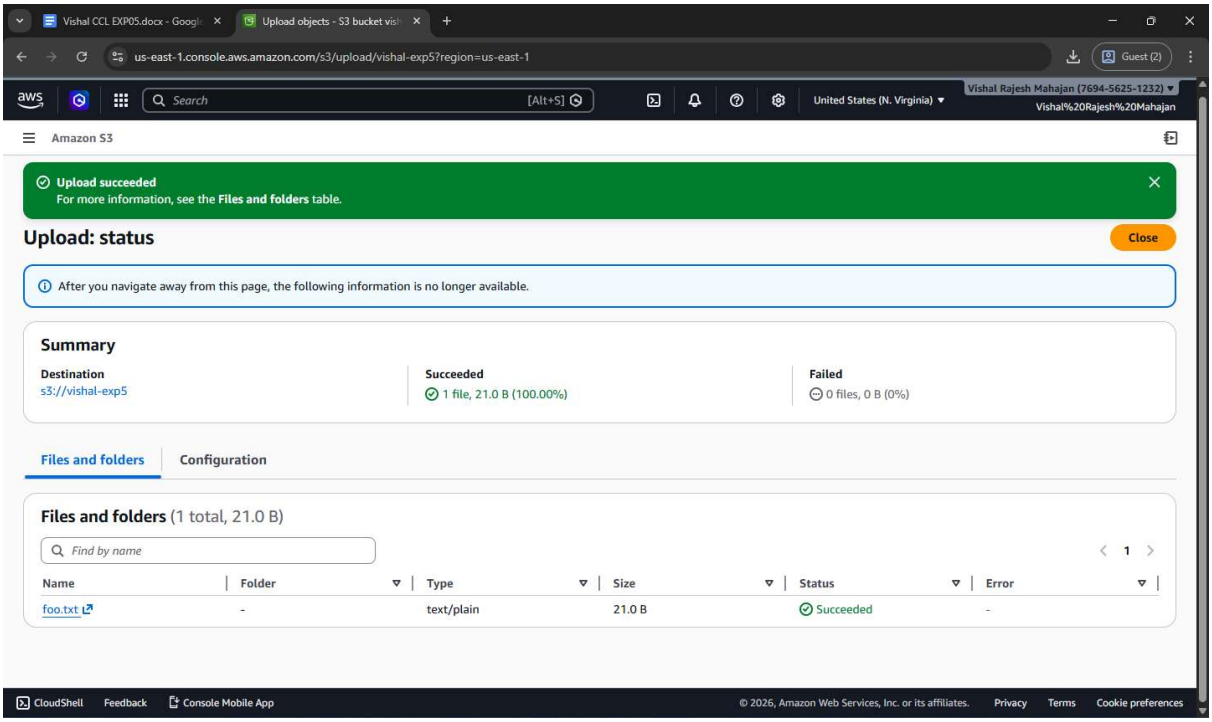



Create Multiple Versions of an Object

3. Create a text file named foo.txt on your computer and write your name in the text file.

4. Save the text file to a location of your choosing.

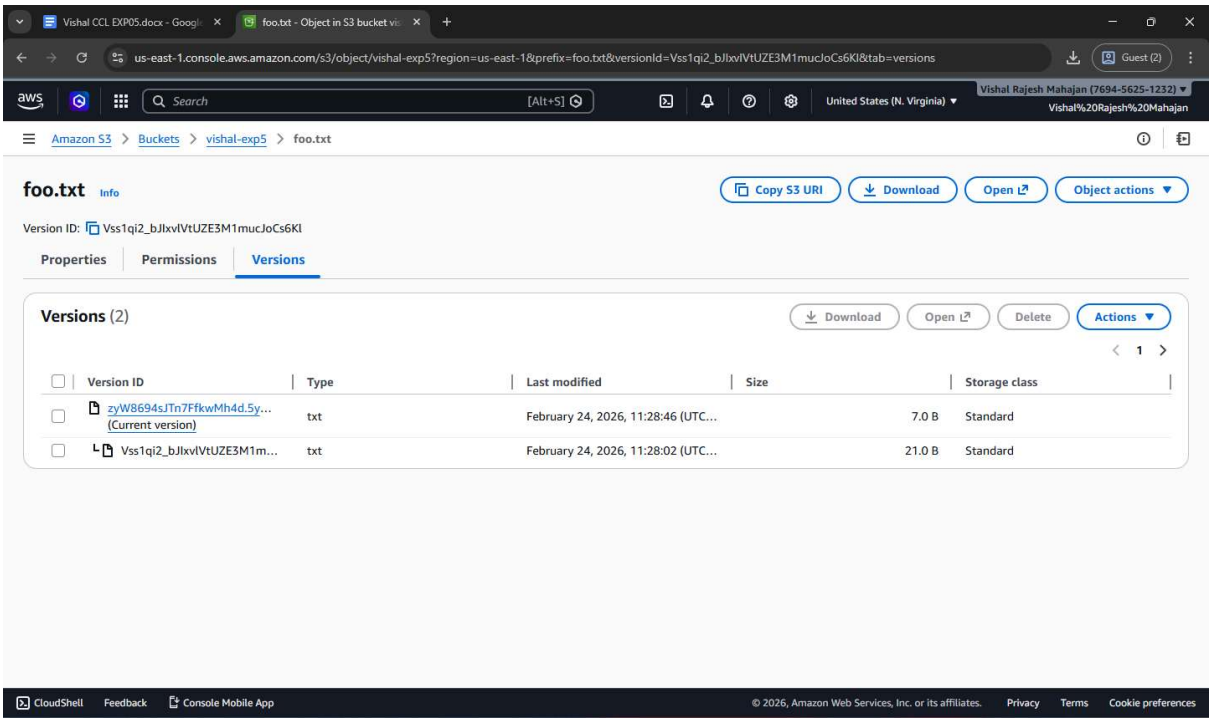
5. Upload the text file to your bucket. This will be version 1.




6. After you have uploaded the text file to your bucket, open the copy on your local computer and change your name to your surname. Save the text file with the original filename.

7. Upload the modified file to your bucket.

8. Select Show Versions on the uploaded object.




**Delete an Object and Then Restore It after versioning**

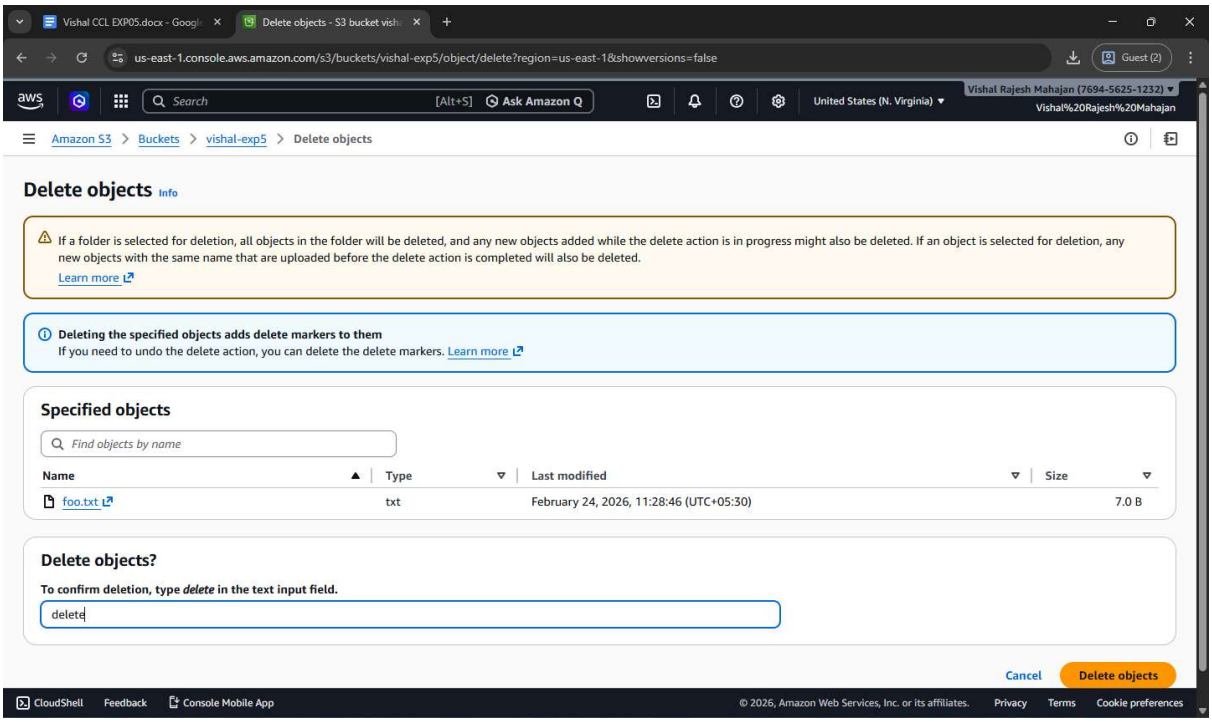
Delete an Object

1. Open the bucket containing the text file for which you now have two versions.

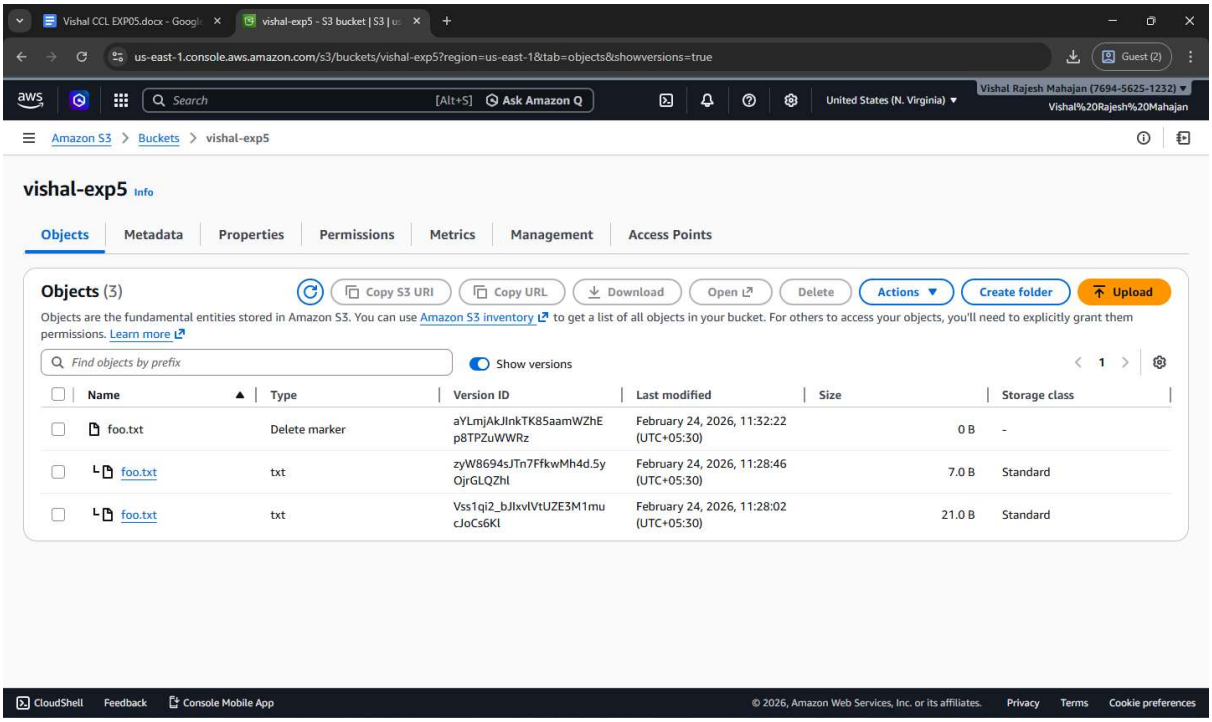
2. Select Hide Versions.

3. Select Delete, and then select OK to verify.

4. Your object will now be deleted, and you can no longer see the object.




5. Select Show Versions. Both versions of the object now show their version IDs.



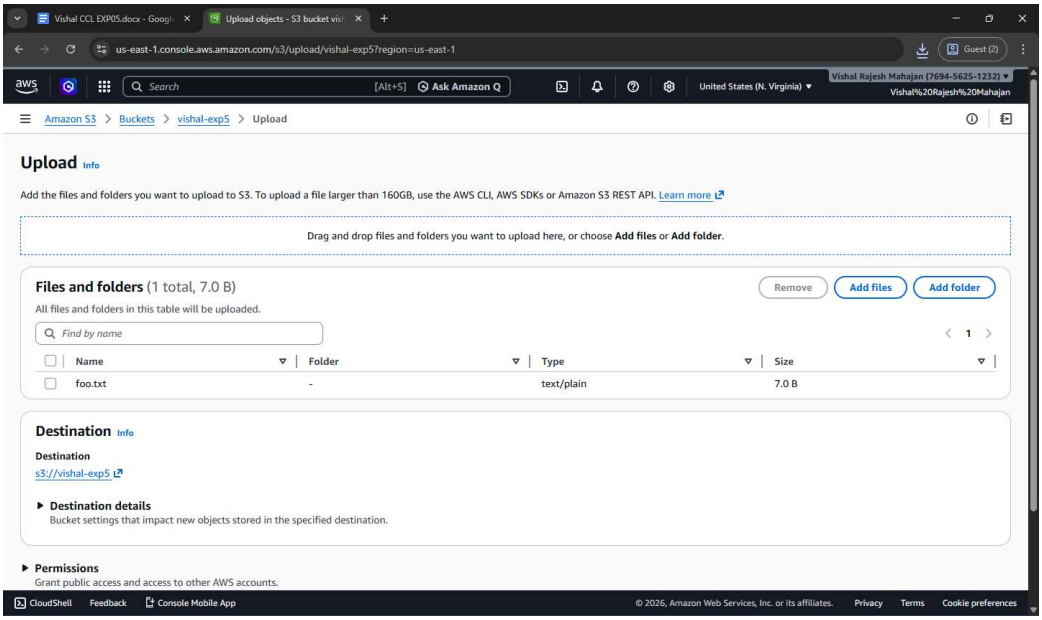
Restore an Object

6. Open your bucket.

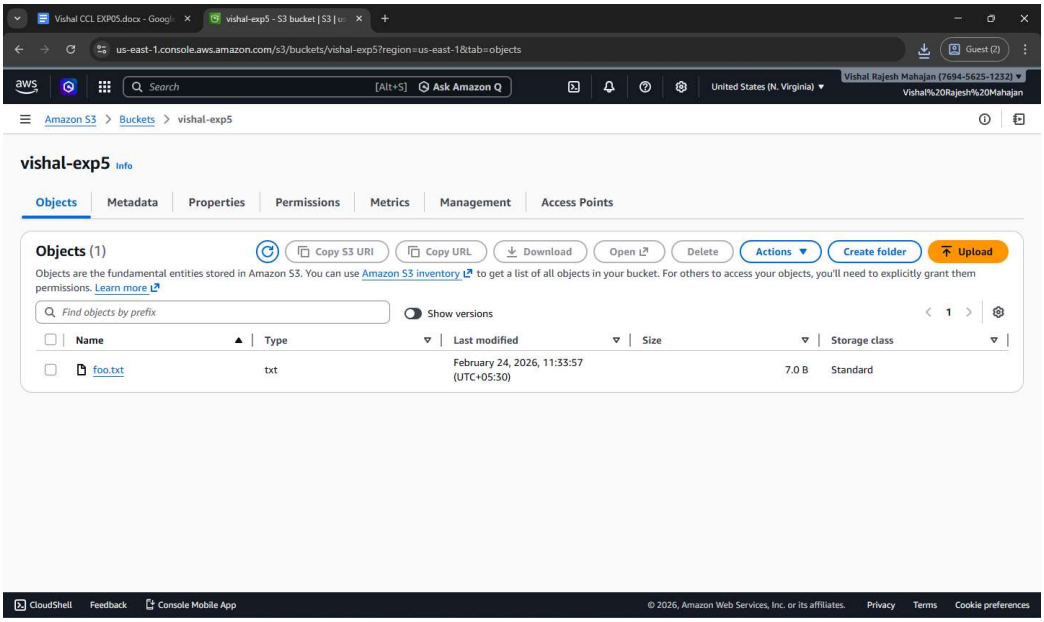
7. Select Show Versions.

8. Select the oldest version and download the object. Note that the filename is simply foo.txt with no version indicator.

9. Upload foo.txt to the same bucket.




10. Select Hide Versions, and the file foo.txt should re-appear.

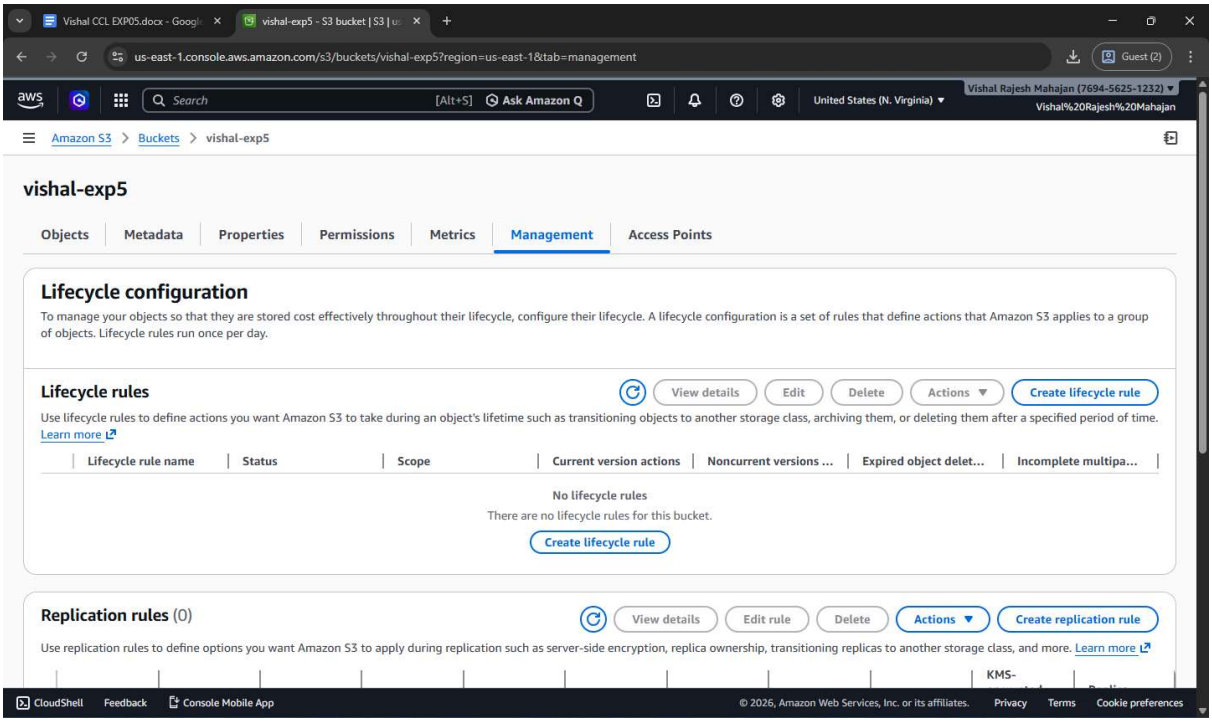


## Lifecycle Management

1. Select your bucket in the Amazon S3 console.

2. Under Properties, add a Lifecycle Rule.

3. Explore the various options to add lifecycle rules to objects in this bucket. It is recommended that you do not implement any of these options, as you may incur additional costs. After you have finished, click the Cancel button.

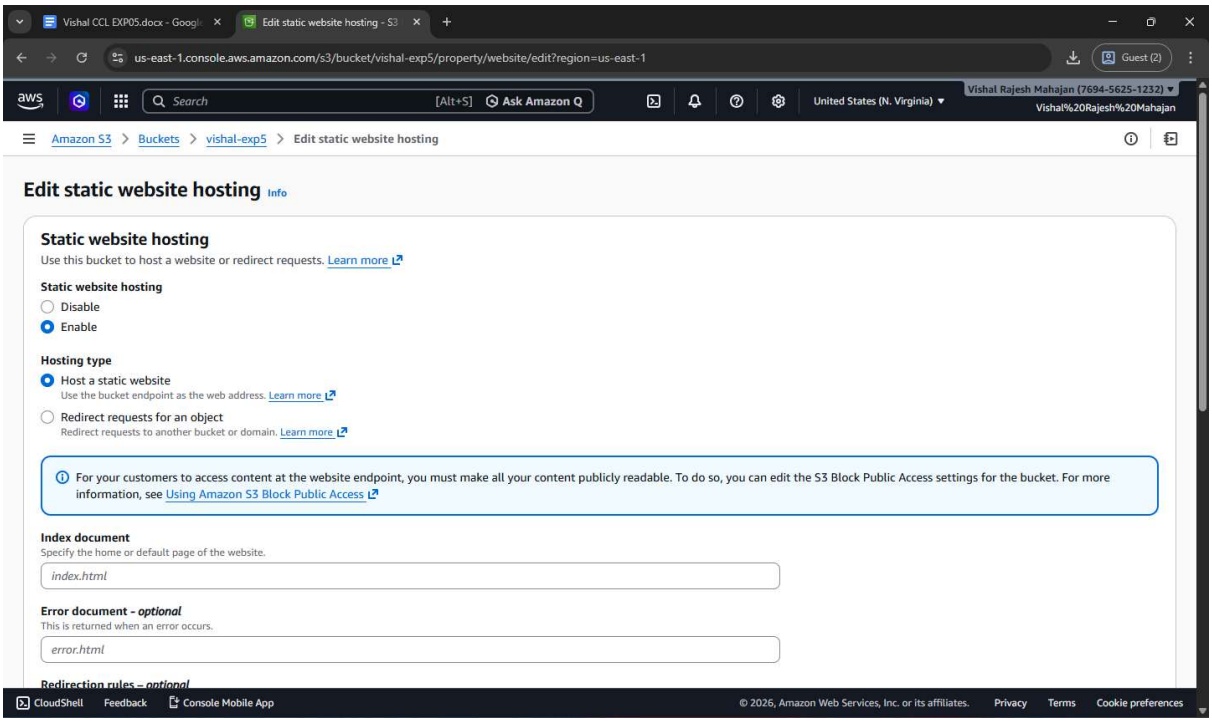



## Post-Experiments Exercise

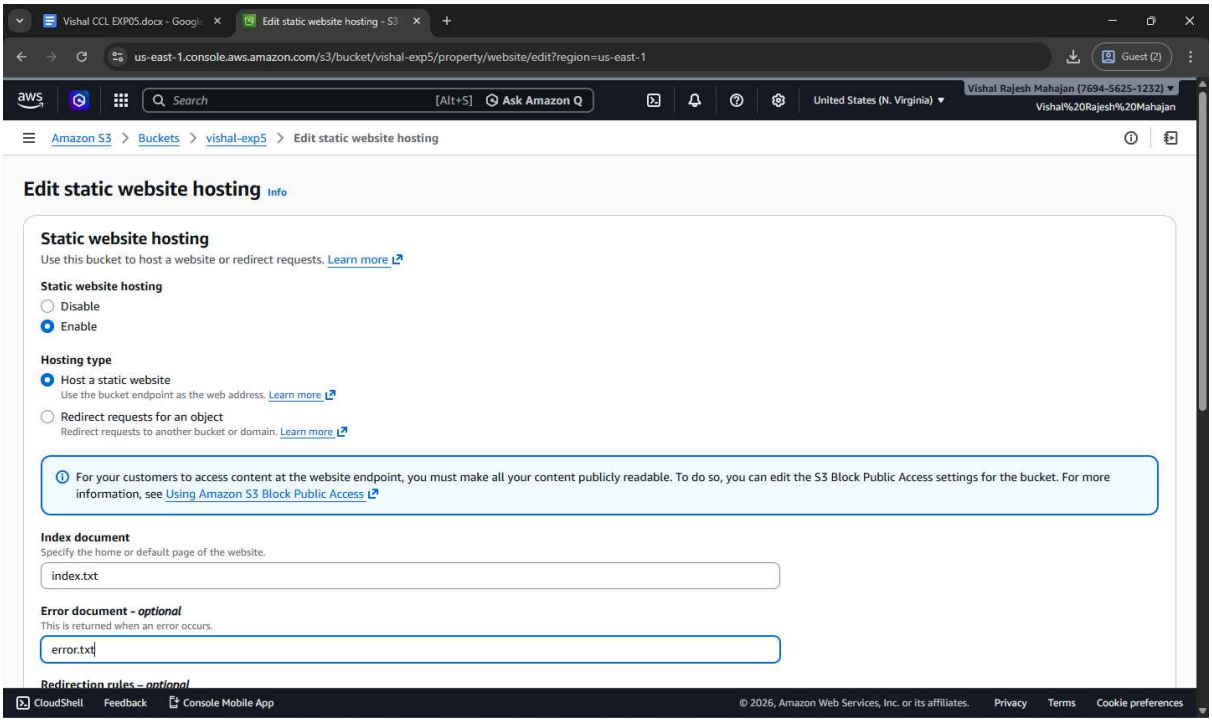
## Enable Static Hosting on Your Bucket

1. Select your bucket in the Amazon S3 console.

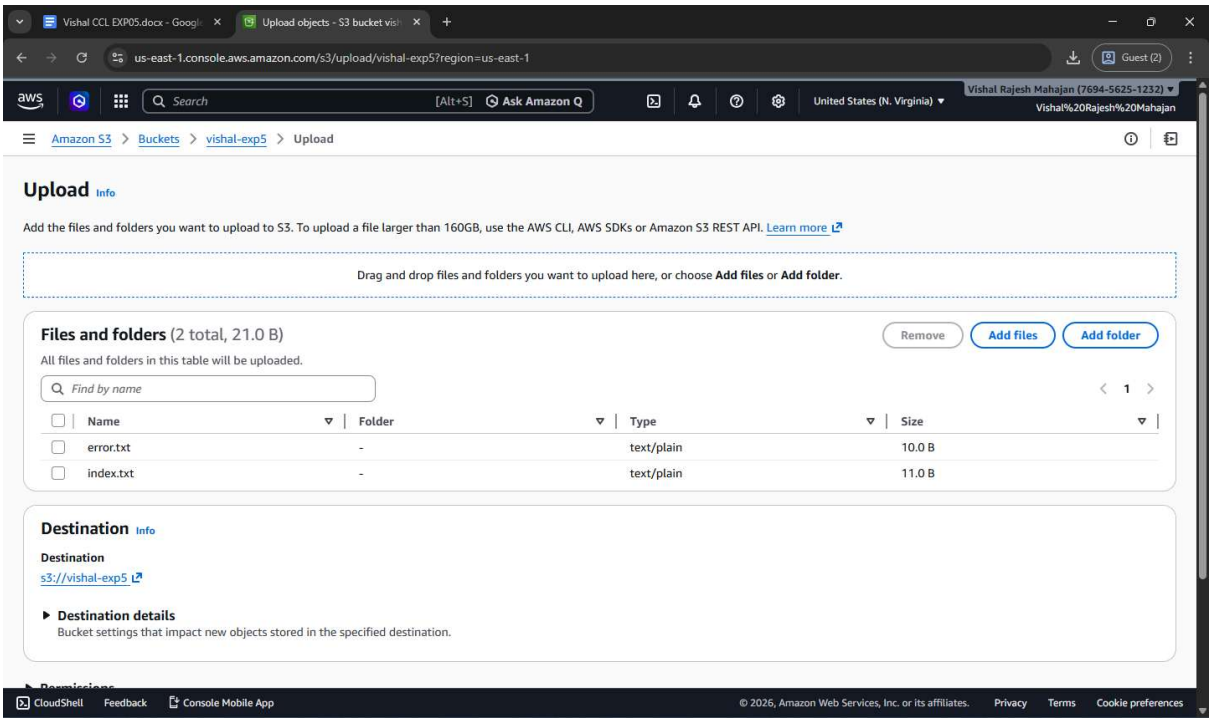
2. In the Properties section, select Enable Website Hosting.



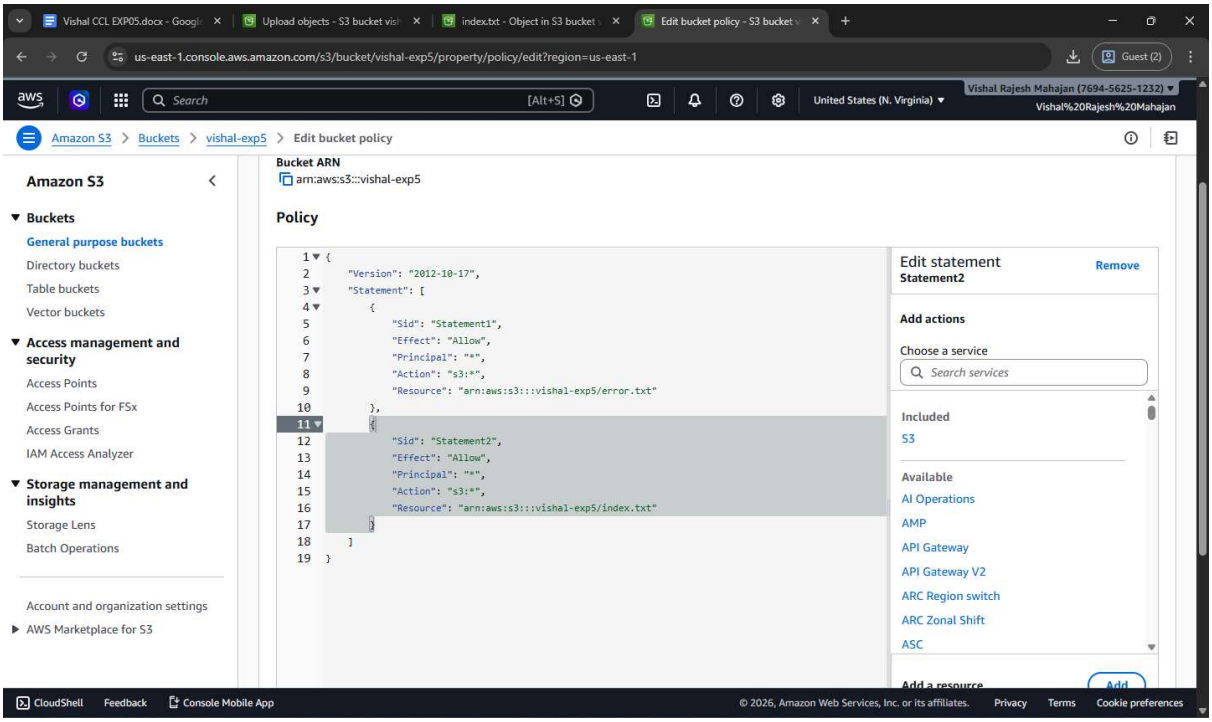

3. For the index document name, enter index.txt, and for the error document name, enter error.txt.



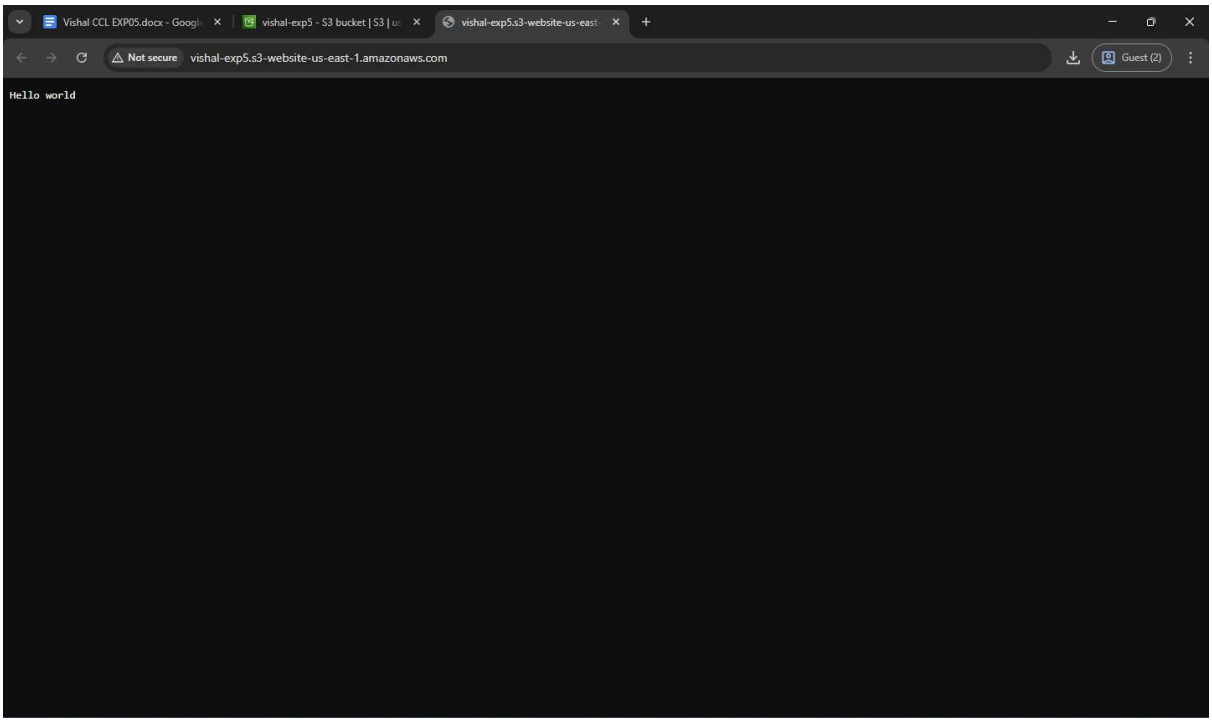

4. Use a text editor to create two text files and save them as index.txt and error.txt. In the index.txt file, write the phrase “Hello World,” and in the error.txt file, write the phrase “Error Page.” Save both text files and upload them to your bucket.



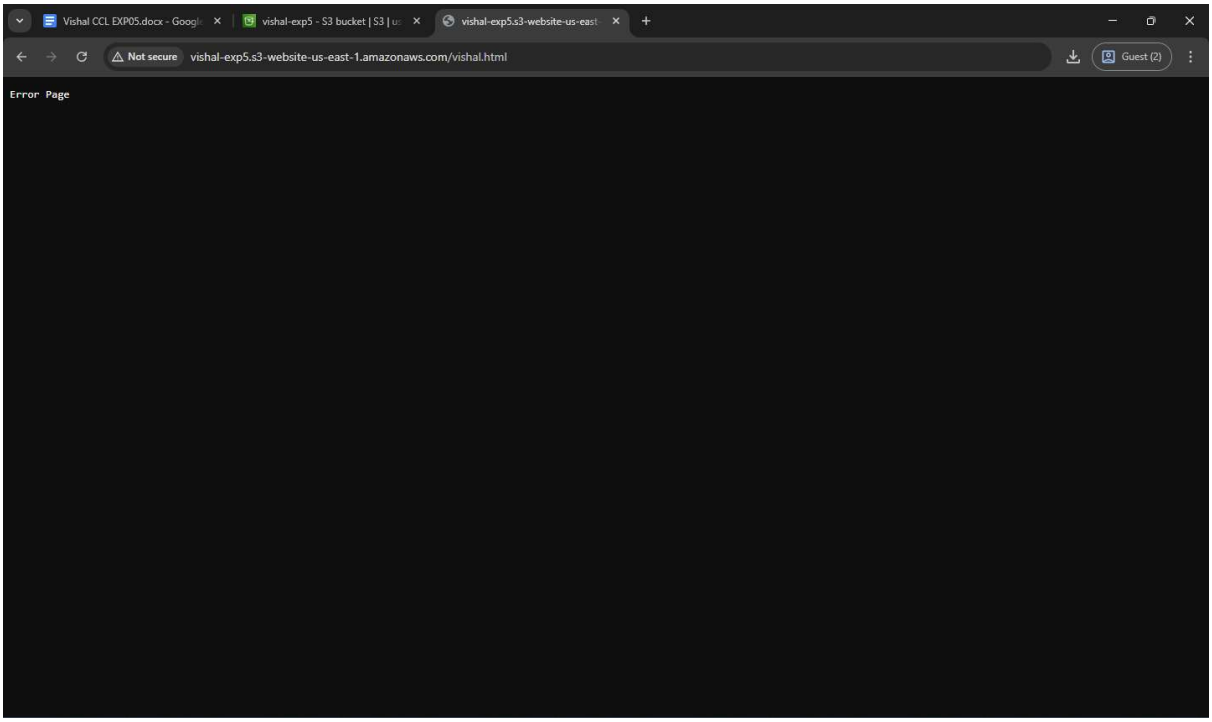

5. Make the two objects public.



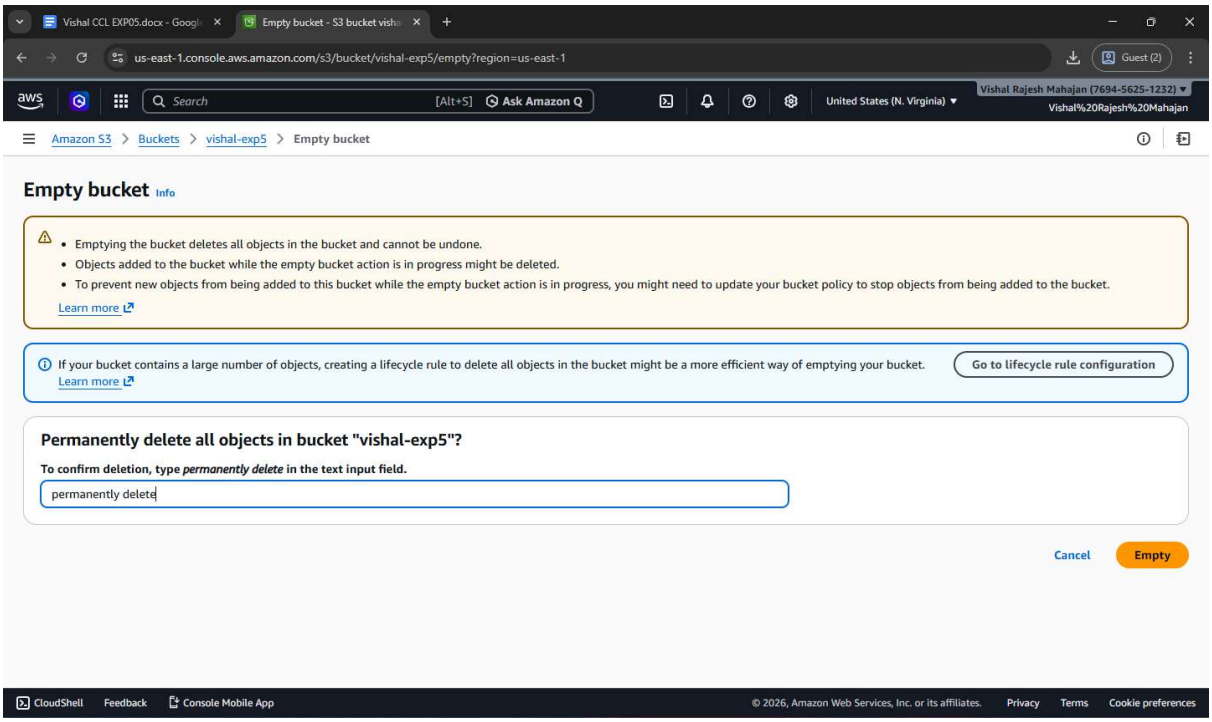

6. Copy the Endpoint: link under Static Website Hosting and paste it in a browser window or tab. You should now see the phrase "Hello World" displayed.

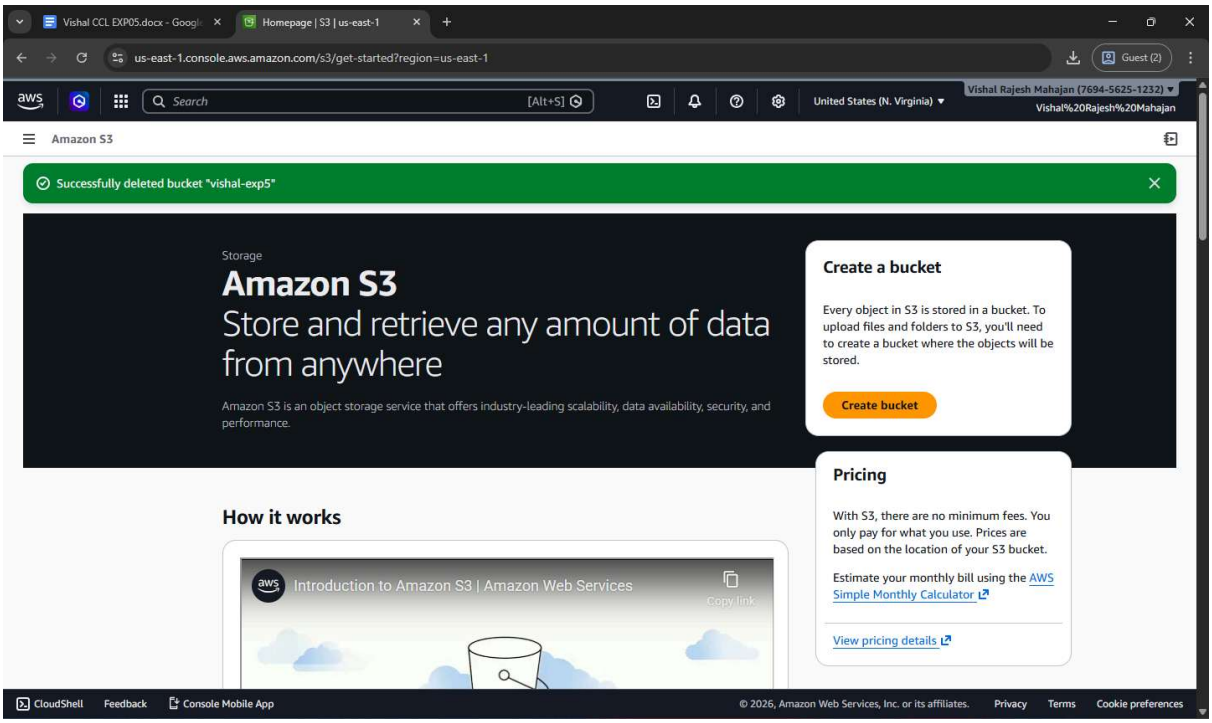



7. In the address bar in your browser, try adding a forward slash followed by a made-up filename (for example, /test.html). You should now see the phrase "Error Page" displayed.




8. To clean up, delete all of the objects in your bucket and then delete the bucket itself.


Vishal CCL EXP05.docx - Google
Empty bucket - S3 bucket visha
us-east-1.console.aws.amazon.com/s3/bucket/vishal-exp5/empty?region=us-east-1
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Vishal Rajesh Mahajan (7694-5625-1232)
Vishal%20Rajesh%20Mahajan
Amazon S3 > Buckets > vishal-exp5 > Empty bucket
Empty bucket Info
Emptying the bucket deletes all objects in the bucket and cannot be undone.
Objects added to the bucket while the empty bucket action is in progress might be deleted.
To prevent new objects from being added to this bucket while the empty bucket action is in progress, you might need to update your bucket policy to stop objects from being added to the bucket.
Learn more
If your bucket contains a large number of objects, creating a lifecycle rule to delete all objects in the bucket might be a more efficient way of emptying your bucket.
Learn more
Go to lifecycle rule configuration
Permanently delete all objects in bucket "vishal-exp5"?
To confirm deletion, type permanently delete in the text input field.
permanently delete
Cancel
Empty
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Vishal CCL EXP05.docx - Google
Homepage | S3 | us-east-1
us-east-1.console.aws.amazon.com/s3/get-started?region=us-east-1
Search
[Alt+S]
United States (N. Virginia)
Vishal Rajesh Mahajan (7694-5625-1232)
Vishal%20Rajesh%20Mahajan
Amazon S3
Successfully deleted bucket "vishal-exp5"
Storage
Amazon S3
Store and retrieve any amount of data from anywhere
Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.
Create a bucket
Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.
Create bucket
Pricing
With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.
Estimate your monthly bill using the AWS Simple Monthly Calculator
View pricing details
How it works
Introduction to Amazon S3 | Amazon Web Services
Copy link
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
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**Extended Theory:**

1. What is storage life cycle and its significance? (to be written in hand)
2. S3 bucket pricing model with example (to be written in hand)
3. Block level storage vs Object level storage(soft copy)

**Results/Calculations/Observations:**

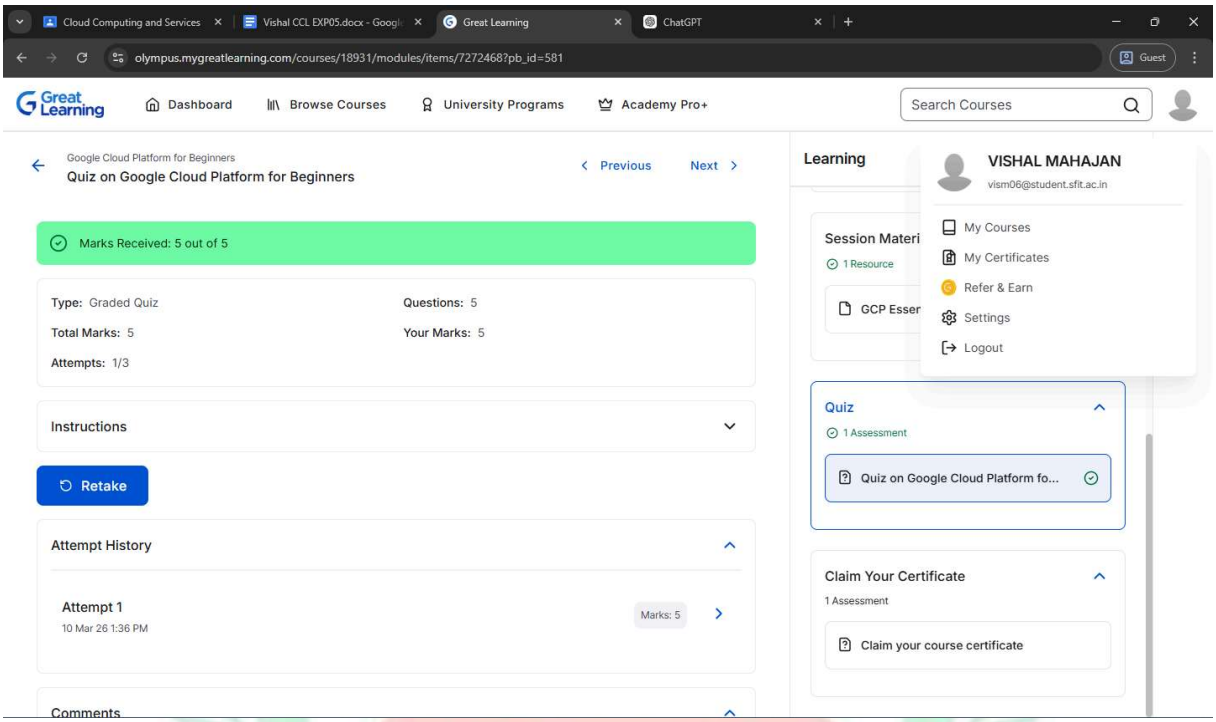
Fill the following observation tables-(to be written in hand)

| Sr. No | | |
|---|---|---|
| 1. | In what ways does Amazon Simple Storage Service (Amazon S3) object storage differ from block and file storage? (Choose 2 answers) | A. Amazon S3 stores data in fixed size blocks.<br>B. Objects are identified by a numbered address.<br>C. Objects can be any size.<br>D. Objects contain both data and metadata.<br>E. Objects are stored in buckets. |
| 2. | Name one more appropriates use case for Amazon Simple Storage Service (Amazon S3)? | 1. Primary storage for a database<br>2. _____ |
| 3. | Your application stores critical data in Amazon Simple Storage Service, which must be protected against inadvertent or intentional deletion. Discuss two solutions? | 1. _____<br>2. _____ |

**Post Experimental Exercise-**

**Questions:**

1. A certificate course on GCP (1Hr)
https://www.mygreatlearning.com/academy/learn-for-free/courses/google-cloud-platform-for-beginners1?utm_source=chatbot&utm_medium=new_flow&utm_campaign=job
Submit test score and certificate of the attended course




**Conclusion:**

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

**References:**

[1] [Online]https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html

[2] [Online]https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary